Zhirui Dai

+1 (858) 200 5325 - zhdai@ucsd.edu - daizhirui.github.io - Google Scholar - LinkedIn - GitHub

Research Interests

Spatiotemporal 3D scene memory · multi-modal neural mapping (SDF, occupancy, radiance, vision-language) · mobile manipulation · safe motion planning · LLM-guided task planning · real-time C++/CUDA acceleration for robot perception and planning

Education

University of California San Diego — **San Diego, CA, USA**
Ph.D. in Electrical and Computer Engineering; Advisor: Prof. Nikolay Atanasov — *01/2022 - Expected 06/2027*

University of California San Diego — **San Diego, CA, USA**
M.S. in Electrical and Computer Engineering — *09/2019 - 06/2021*

Fudan University — **Shanghai, China**
B.S. in Physics — *09/2015 - 06/2019*

Skills & Open Source

Languages: C++, CUDA, Python, C, MATLAB, Java, Swift
Libraries & Tools: PyTorch, ROS, OpenCV, Open3D, Docker, Kubernetes, LangChain, ManiSkill
Open Source Contributions: PyTorch, Open3D, PointCloudLibrary, pybind11

Research Experience

Research Intern, VinMotion Inc. — **Irvine, CA, USA**
Real-Time Multi-Modal 3D Mapping for Robots — *06/2026 - 09/2026*

- **OREN-X: Octree Residual Network for Real-Time Multi-Modal Mapping** — *Submitted to ICRA 2027*
  - Developed an online mapping system that jointly learns **SDF, occupancy, radiance and vision-language fields** in a single octree, running at **80+ fps** for SDF and **30+ fps** for all four modalities.
  - Designed multi-pool octree indexing (per-modality resolution) and **cross-modality supervision**, improving near-surface SDF accuracy by **33%** over single-modality baselines and open-vocabulary 3D mIoU by **71%** over prior methods.
  - Compressed vision-language features **3.7x** with online dictionary learning and built GPU-accelerated ray-octree traversal for real-time rendering; backbone-agnostic (CLIP, TIPS).
  - Validated on Replica and TUM RGB-D and on real-world data from a **Jackal robot** (lab and construction site) and a **humanoid**.

Graduate Student Researcher, UCSD Existential Robotics Lab — **San Diego, CA, USA**
Advisor: Prof. Nikolay Atanasov — *01/2022 - Present*

- **Vision-Language 3D Mapping for Mobile Manipulation** — *ICRA 2026, RSS 2025 WS*
  - Architected SBP, an incremental 3D language-feature mapping framework that gives mobile manipulators **spatiotemporal memory** in unseen environments, using **hierarchical hash-grid maps** to compress **2D VLM features (CLIP, DINO)** with a **cross-scene generalized decoder**.
  - Developed a **transformer-based manipulation policy** conditioned on the 3D map, outperforming image-based policies in in-distribution and novel scenes (up to **15% higher success rate**).
  - Engineered **ManiSkill-HAB** data pipelines to synthesize large-scale multi-modal datasets (RGB-D, semantics, task descriptions) for training.
- **Neural Signed Directional Distance Functions (SDDF)** — *TPAMI 2026, RSS 2025 WS Best Paper*
  - Introduced SDDF, a 3D representation enabling single-pass occlusion reasoning and collision checking; a hybrid model of ellipsoidal priors and neural residuals achieves **SOTA reconstruction accuracy** on Replica.
  - Wrote custom CUDA kernels for ellipsoidal SDDF, indexed pose transforms and indexed linear layers: up to **600x speedup** and **50–90% GPU memory savings** over PyTorch.
  - Deployed on aerial platforms for real-time sensor view coverage maximization and collision avoidance.
- **Real-Time SDF Mapping & Motion Planning** — *IROS 2026, RA-L 2026, IJRR 2025*
  - Proposed OREN, an octree residual network for globally accurate real-time SDF learning from LiDAR/RGB-D, beating SOTA in accuracy and efficiency (**14 fps** desktop GPU, **7 fps on Jetson Orin**).
  - Open-sourced a semi-sparse octree library (C++/CUDA; up to **550x** faster voxel indexing) and Kernel-SDF, a C++ kernel regression library for SDF estimation with **uncertainty quantification** and Python/**ROS** interfaces.
  - Co-designed distance-accelerated motion planning (Bubble*) running onboard a quadrotor, planning ≈90 m trajectories in **1–3 s** vs. up to 10 s for baselines.
- **LLM-Guided Robot Task Planning** — *ICRA 2024, IROS 2025*
  - Built an optimal scene graph planner that translates natural language into LTL via **LangChain** and uses LLM heuristics for a **1600x speedup** over AMRA* with guaranteed optimality.
  - Co-developed LTLCodeGen, generating **100% syntactically correct** LTL specifications and outperforming fine-tuned baselines.

Graduate Student Researcher, UCSD Statistical Visual Computing Lab — **San Diego, CA, USA**
Advisor: Prof. Nuno Vasconcelos — *06/2020 - 08/2021*

- **BEV-Net: Social Distance Monitoring from Surveillance Video** — *ICCV 2021*
  - Developed BEV-Net with a novel **Bird's-Eye-View projection layer** and calibration-free camera pose recovery from human keypoints, achieving **30% lower** distance error and **2x higher IoU** than SOTA.

Industry Experience

Research Intern — VR/AR Systems & Deep Learning **08/2021 - 12/2021**

XCOM Labs Inc., CA, USA

- Built a neural 6-DoF head & hand pose predictor for multi-user VR/AR remote rendering, with **90% lower** trajectory error than the OpenVR baseline, reducing motion-to-photon latency; curated the proprietary dataset used for benchmarking.

Software Development Intern — SoC Software Development **01/2018 - 06/2019**

Camel Microelectronics Inc., CA, USA

- Architected CamelStudioX, an IDE for low-power SoC development, and built the embedded stack (HAL, soft-float library, cross-compiler toolchain, ELF tools).

Publications

*: equal contribution.

Journal & Conference Papers

- Z. Dai*, T. Fan*, M. Amani, J. Seo, K. M. B. Lee, H. Oh, N. Atanasov. "**Kernel-SDF: An Open-Source Library for Real-Time Signed Distance Function Estimation using Kernel Regression.**" IEEE Robotics and Automation Letters (RA-L), 2026. Paper Project
- Z. Dai*, Q. Qian*, T. Fan, N. Atanasov. "**OREN: Octree Residual Network for Real-Time Euclidean Signed Distance Mapping.**" IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. Paper Project
- T. Matthews, M. Kumar A. R., A. Begzadić, S. Tonkens, Z. Dai, N. Atanasov, J. Cortés, S. Herbert. "**Expanding Safe Sets with Learning-Based Barrier Functions and Reachability.**" IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. Paper
- Z. Dai, H. Shin, Y. Tian, K. M. B. Lee, N. Atanasov. "**Learning Scene-Level Signed Directional Distance Function with Ellipsoidal Priors and Neural Residuals.**" IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026. Paper Project
- S. Kim, W. Chung, Z. Dai, D. Bhatt, A. Shukla, H. Su, Y. Tian, N. Atanasov. "**Seeing the Bigger Picture: 3D Latent Mapping for Mobile Manipulation Policy Learning.**" IEEE International Conference on Robotics and Automation (ICRA), 2026. Paper Project
- B. Rabiei, M. Kumar A. R., Z. Dai, S. L. S. R. Pilla, Q. Dong, N. Atanasov. "**LTLCodeGen: Code Generation of Syntactically Correct Temporal Logic for Robot Task Planning.**" IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025. Paper Project
- K. Long, Y. Yi, Z. Dai, S. Herbert, J. Cortés, N. Atanasov. "**Sensor-Based Distributionally Robust Control for Safe Robot Navigation in Dynamic Environments.**" The International Journal of Robotics Research (IJRR), 2025. Paper Project
- Z. Dai, A. Asgharivaskasi, T. Duong, S. Lin, M.-E. Tzes, G. J. Pappas, N. Atanasov. "**Optimal Scene Graph Planning with Large Language Model Guidance.**" IEEE International Conference on Robotics and Automation (ICRA), 2024. Paper Project
- Z. Dai, Y. Jiang, Y. Li, B. Liu, A. B. Chan, N. Vasconcelos. "**BEV-Net: Assessing Social Distancing Compliance by Joint People Localization and Geometric Reasoning.**" IEEE/CVF International Conference on Computer Vision (ICCV), 2021. Paper

Preprints

- M. Amani*, M. Alyaseen*, Z. Dai, K. M. B. Lee, N. Atanasov, J. Cortés. "**BubbleCBF: Constructive Safety for Second-Order Systems and Application to Autonomous Robot Navigation in Unknown Environments.**" Under review, 2026. Paper Project
- Z. Dai, Q. Qian, D. M. Nguyen, Q.-D. Pham, K. Bronder, C. Nieto-Granda, Y. Chen, Q. Nguyen, N. Atanasov. "**OREN-X: Octree Residual Network for Real-Time Multi-Modal Mapping.**" Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027, under review. Paper Project
- Q.-D. Pham, A. Dao, D. V. Le, N. V. T. Pham, T. A. Nguyen, Z. Dai, Y. Chen, T. P. Le, V.-T. Nguyen, Q. Nguyen. "**AdaGeoVLN: Selective Geometry Across Representation Depth and Navigation Time for Vision-Language Navigation.**" Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027, under review. Paper
- J. Stanley*, Z. Dai*, Q. Qian, T.-C. Ho, T. Fan, S. Saha, C. Barngrover, K. M. B. Lee, N. Atanasov. "**From Distances to Trajectories: Real-Time Signed Distance Function Mapping and Distance-Accelerated Motion Planning for UAVs.**" Submitted to Autonomous Robots, 2026, under review. Paper Project
- K. M. B. Lee, Z. Dai, C. Le Gentil, L. Wu, N. Atanasov, T. Vidal-Calleja. "**Safe Bubble Cover for Motion Planning on Distance Fields.**" arXiv preprint arXiv:2408.13377, 2024. Paper

Workshop Papers

- J. Stanley*, Z. Dai*, T.-C. Ho, S. Saha, C. Barngrover, N. Atanasov. "**From Distances to Trajectories: Real-Time Signed Distance Function Mapping and Distance-Accelerated Motion Planning for UAVs.**" ICRA 2026 Workshop on Aerial Inspection for Marine Infrastructures (AIMI), 2026. Paper Project
- Z. Dai, H. Shin, Y. Tian, K. M. B. Lee, N. Atanasov. "**Learning Scene-Level Signed Directional Distance Function for Aerial Autonomy.**" Workshop on Leveraging Implicit Methods for Aerial Autonomy at RSS, 2025. Paper Project **[Best Paper Award]**
- S. Kim, W. Chung, Y. Tian, Z. Dai, A. Shukla, H. Su, N. Atanasov. "**Seeing the Bigger Picture: 3D Latent Mapping for Mobile Manipulation Policy Learning.**" Workshop on Mobile Manipulation and Workshop on Learned Robot Representations at RSS, 2025. Paper Project **[Best Paper Nomination]**

Awards & Honors

Best Paper Award, RSS 2025 Workshop on Leveraging Implicit Methods for Aerial Autonomy 2025
Best Paper Nomination, RSS 2025 Workshops on Mobile Manipulation & Learned Robot Representations 2025
Electrical and Computer Engineering Department Fellowship, UCSD 2022
Outstanding Graduate, Fudan University 2019

Service & Teaching

Reviewer: NeurIPS, CVPR, RSS, CoRL, ICRA, IROS, RA-L, T-ITS
Teaching Assistant, ECE276B: Planning & Learning in Robotics, UCSD Spring 2023, Spring 2024
Mentor, GEAR, ENLACE, SRIP, STARS undergraduate research programs 2021 - 2022